

MEDIGUARD

Clinical Prescription Verification System

Department of Computer Science · SZABIST University, Karachi
Finite Automata & Formal Languages · 2026

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COURSE DETAILS

Finite Automata & Formal Languages

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[Github](#)

01 PROJECT OVERVIEW

MediGuard is a C application that automates the safety screening of medical prescriptions before any drug is dispensed to a patient. Built on the theoretical foundation of Finite State Machines (FSMs) from formal language theory, the system processes each prescription through a structured, sequential pipeline of validation checks mimicking the multi-step review process that clinical pharmacists perform manually.

The project demonstrates how a concept from theoretical computer science, the FSM, can be directly translated into a real, working safety-critical application in the healthcare domain.

Two Modes of Operation

Interactive Mode: The pharmacist enters prescription data step by step. If any check fails, the system prompts for correction rather than halting abruptly.

Automated Batch Mode: Five pre-loaded test cases run automatically to showcase all fraud and error detection capabilities in sequence.

Medication errors are among the most common and preventable causes of patient harm in healthcare settings worldwide. Errors in prescription processing , wrong drug, incorrect dose, dangerous combinations, undetected allergies can have severe or fatal consequences. In many hospitals, these checks are still performed manually by pharmacists, making them prone to human fatigue and oversight.

MediGuard addresses this problem by automating the verification pipeline using a formal computational model. A Finite State Machine is ideal for this use case because:

- › Each check is discrete and independent : perfectly modeled as a state.
- › The machine can only be in one state at a time : preventing partial or ambiguous checks.
- › Transitions are deterministic : the same input always produces the same outcome.
- › Rejection is immediate : as soon as one check fails, the machine halts with a clear reason.

3.1 The State Machine — All States Explained

The FSM has 10 primary states plus retry states for interactive mode. The machine begins at Q0 and either reaches Q_ACCEPT (prescription cleared) .

State	Name	Description
Q0	Start	Initial state. No processing occurs here , transitions immediately to ID check.
Q_CHECK_ID	ID Verification	Parses and validates the 9-digit combined ID. Checks year, category, serial number, and department code.
Q_LOOKUP_DRUG1	Primary Drug Lookup	Searches the formulary database for the primary drug. Verifies department match and stock availability.
Q_CHECK_DOSE1	Primary Dose Validation	Compares the prescribed dose against the safe limit. Automatically applies a 50% reduction for pediatric patients under 20 kg.
Q_LOOKUP_DRUG2	Secondary Drug Lookup	Same as Drug 1 but for an optional second drug. This state is skipped entirely if no secondary drug is prescribed.
Q_CHECK_DOSE2	Secondary Dose Validation	Same as Dose 1 but for the secondary drug. Pediatric halving applies here too.
Q_CHECK_ALLERGY	Allergy Screening	Cross-references the patient's allergy disclosures against both drugs. Enforces cross-reactivity rules (e.g., penicillin blocks amoxicillin).
Q_CHECK_INTERACTION	Drug Interaction Check	Blocks two specific drug pairs known to cause dangerous adverse effects when combined.
Q_ACCEPT	Approved	Final accepting state. All checks passed. The prescription is safe to dispense.

3.2 The Internal Drug Formulary

The system maintains an in-memory database of 4 approved drugs. Each entry stores the drug code, its department association, the maximum clinically safe dose for an adult patient, and current inventory stock:

Drug Code	Associated Dept.	Max Adult Dose	Current Stock
AMOX500	General Medicine (Dept. 12)	1,000 mg	50 units — Available
PENC250	General Medicine (Dept. 12)	500 mg	0 units — OUT OF STOCK
IBU200	Orthopedics (Dept. 14)	800 mg	120 units — Available
WARF5	Orthopedics (Dept. 14)	500 mg	80 units — Available

3.3 The 9-Digit Combined Prescription ID

Every prescription is identified by a 9-digit number that encodes both patient information and prescription metadata. The system extracts and validates each segment independently:

Position	Field Name	Format / Constraint
Digits 1–2	Enrollment year	Must equal '26' (year 2026 only — older IDs are rejected)
Digit 3	Patient category	1 = Emergency admission, 2 = OPD (outpatient)
Digits 4–5	Patient serial number	Numeric value between 01 and 99
Digits 6–7	Department code	12 = General Medicine, 14 = Orthopedics
Digits 8–9	Medical record number	Unique patient record identifier within the department

04 THE FIVE-STAGE VALIDATION PIPELINE

Every prescription passes through five sequential validation stages. A single failure at any stage stops the pipeline immediately and generates a rejection report.

Stage 1: Patient & Prescription ID Verification

The 9-digit combined ID undergoes four sub-checks: (1) length must be exactly 9 characters, (2) all characters must be numeric, (3) the year prefix must be '26', (4) the category digit must be 1 or 2, (5) the serial number must fall in the range 01–99, and (6) the department code extracted from positions 6–7 must correspond to a registered department in the formulary.

Why is the year locked to 2026?

The system is designed for the current academic and clinical batch. Locking the year prevents old or recycled prescription IDs from being reused, which is a common vector for prescription fraud.

Stage 2 : Drug Formulary Verification

For each prescribed drug (primary and secondary), the system performs three independent checks using substring matching against the formulary:

- › Existence check : the drug code must appear in the database. Unknown drugs are immediately rejected.
- › Department match : the drug's registered department must match the department encoded in the prescription ID. A General Medicine drug cannot be dispensed on an Orthopedics prescription, and vice versa.
- › Stock availability : the drug's stock count must be greater than zero. A drug that is listed in the formulary but has zero stock is still blocked from dispensing.

Stage 3 — Dosage Safety Validation

The prescribed dose is compared against the maximum safe threshold stored in the formulary. The system applies a special rule for pediatric patients:

Patient Profile	Weight Condition	How the Dose Limit is Applied
Adult patient	Weight $\geq$ 20 kg	Full maximum dose from the formulary is used as the safety threshold.
Pediatric patient	Weight $<$ 20 kg	The maximum dose is automatically halved before comparison. This prevents accidental overdose in low-weight patients who have a lower metabolic tolerance.

Stage 4 — Allergy Cross-Reference Screening

The patient's allergy disclosure string is split by commas and each allergen token is checked against both prescribed drug codes using substring matching. This means partial matches are caught, for example, a patient who lists 'AMOX' as an allergen will be blocked from receiving 'AMOX500'.

The system also enforces a pharmacological cross-reactivity rule: if the patient lists 'PENC' or 'PENICILLIN' as an allergen, Amoxicillin (any drug code containing 'AMOX') is also automatically blocked. This is because both drugs belong to the beta-lactam antibiotic family, and patients allergic to one frequently react to the other.

Stage 5 — Adverse Drug Interaction Detection

Even when two drugs are individually safe, certain combinations cause dangerous physiological reactions. The system permanently blocks two specific pairs:

Blocked Combination	Clinical Reason
IBU200 + WARF5 (Ibuprofen + Warfarin)	Ibuprofen inhibits platelet function and can displace warfarin from plasma proteins, dramatically increasing warfarin's blood-thinning effect. The combination creates a high risk of uncontrolled internal bleeding and haemorrhage.
PENC250 + AMOX500 (Penicillin G + Amoxicillin)	Both drugs share the same mechanism of action — inhibiting bacterial cell wall synthesis via the same binding site. Prescribing both simultaneously offers no additional benefit and is medically redundant. It also increases the risk of side effects without therapeutic gain.

05 TEST CASES & EXPECTED OUTCOMES

In Automated Batch Mode, the system processes five pre-defined prescriptions. Each is designed to test a specific validation rule:

#	Prescription Input	Result	Reason
01	ID: 261051488 Drug: IBU200 400mg Weight: 70kg Allergy: NONE	APPROVED	Every check passes. Valid ID, drug in stock, dose within 800mg limit, no allergy, no dangerous interaction.
02	ID: 251051488 Drug: IBU200 400mg Weight: 70kg Allergy: NONE	REJECTED	ID starts with '25' indicating year 2025. The system only accepts 2026 IDs. Rejected at Stage 1.
03	ID: 261051288 Drug: PENC250 250mg Weight: 65kg Allergy: NONE	REJECTED	PENC250 has zero stock. Even though the drug is registered and the dose is safe, no stock means no dispensing. Rejected at Stage 2.
04	ID: 261051288 Drug: AMOX500 650mg Weight: 14kg Allergy: NONE	REJECTED	Patient weighs 14kg (under 20kg threshold). The adult max of 1000mg is halved to 500mg. The prescribed 650mg exceeds this pediatric limit. Rejected at Stage 3.
05	ID: 261051488 Drug 1: IBU200 200mg Drug 2: WARF5 150mg Weight: 75kg Allergy: NONE	REJECTED	Both drugs individually pass all checks — valid, in stock, doses safe. But their combination is a blocked interaction due to severe internal bleeding risk. Rejected at Stage 5.

One of MediGuard's key design features is its graceful error handling in Interactive Mode. Rather than crashing or giving a simple rejection, the system uses retry states (e.g., Q_RETRY_ID, Q_RETRY_DRUG1) that move the tape head back and prompt the user to re-enter the problematic field.

This mirrors how a real pharmacist workflow operates . a single data entry error should not force the operator to restart the entire prescription from scratch. The retry mechanism makes the system far more practical for real clinical environments.

Tape Head Metaphor

In Turing Machine theory, moving the 'tape head left' means going back to re-read and rewrite a symbol. In MediGuard, this concept is implemented as moving back to a retry state reprocessing only the failed field while preserving all previously validated fields.

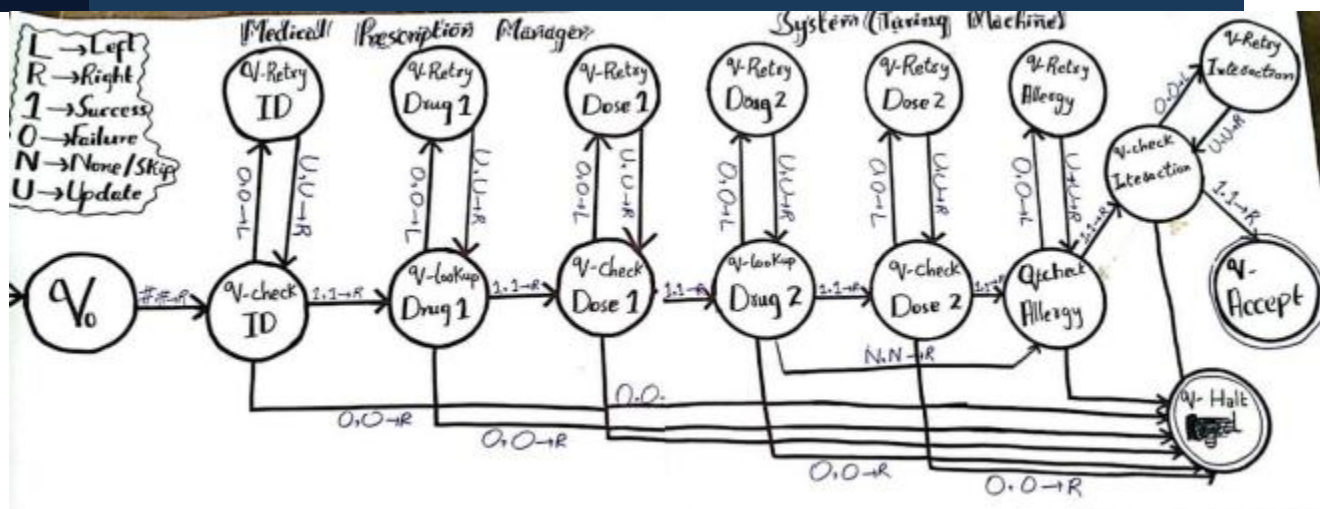
In Batch Mode, no retries are attempted. The machine halts immediately on the first failure and displays the rejection reason. This is intentional — batch processing assumes inputs are final and any failure must be logged and reviewed manually.

07 REAL-WORLD APPLICABILITY

While MediGuard is a prototype built for academic demonstration, its architecture closely mirrors real clinical decision support systems (CDSS) used in hospitals. The following table maps MediGuard's features to their real-world healthcare equivalents:

MediGuard Feature	Real-World Equivalent
9-digit prescription ID format	Hospital Management System (HMS) prescription numbering with department and year encoding
Drug formulary database	Hospital's approved drug list typically managed by a clinical pharmacist committee
Stock availability check	Real-time pharmacy inventory system integration
Pediatric dose halving	Weight-based dosing protocols used in paediatric wards (mg/kg calculations)
Allergy cross-reactivity (Penicillin → Amoxicillin)	Drug allergy databases like First Databank or Medi-Span that track drug family cross-sensitivities
Drug interaction detection	Clinical drug interaction databases (e.g., Drugs.com, Micromedex) used in hospital formulary systems
Interactive retry states	Electronic prescribing systems that flag errors inline and allow immediate correction
Batch mode auto-testing	Automated regression testing used in healthcare software validation (IEC 62304 compliance)

08 TURING MACHINE



MediGuard demonstrates that formal computational models specifically Finite State Machines are not merely academic abstractions. They can be directly implemented as reliable, deterministic safety systems in high-stakes real-world domains like healthcare.

The system's five-stage validation pipeline covers the most common and dangerous classes of prescription errors: invalid identification, unlicensed or out-of-stock drugs, overdose (with automatic pediatric protection), patient allergy conflicts (including cross-reactive drug families), and life-threatening drug interactions.

Each stage maps cleanly to a formal FSM state, making the system's logic transparent, auditable, and extensible. Future enhancements could include a larger drug database, real-time stock API integration, expanded drug interaction pairs, and a graphical interface for clinical staff.

GitHub Repository

https://github.com/Ayesha-bilal/Medical_Safety_Automata_System — Full source code, test cases, and documentation available.